| Reg. No. | | | | | | | | | |
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# MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

*(A constituent unit of MAHE, Manipal)*

## VII SEMESTER B. TECH (ELECTRICAL & ELECTRONICS ENGINEERING)
## END SEMESTER EXAMINATIONS, NOVEMBER-DECEMBER 2024

## MODERN POWER CONVERTERS [ELE 4085]

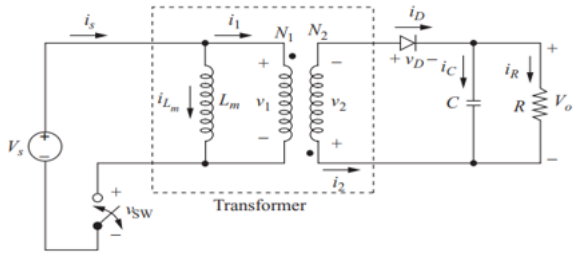
REVISED CREDIT SYSTEM

**Time: 3 Hours** **Date: 25 NOVEMBER 2024** **Max. Marks: 50**

**Instructions to Candidates:**

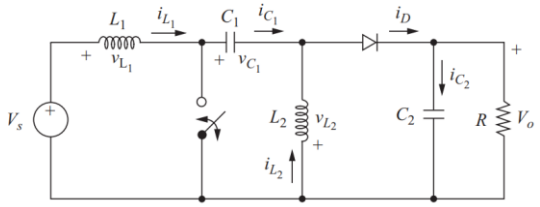
- Answer **ALL** the questions.
- Missing data may be suitably assumed.

**1A.** The output voltage expression is $V_o = V_s\left(\frac{D}{1-D}\right)\left(\frac{N_2}{N_1}\right)$ for the DC-DC converter given below operating in continuous conduction mode with duty cycle D. Justify using appropriate expressions.



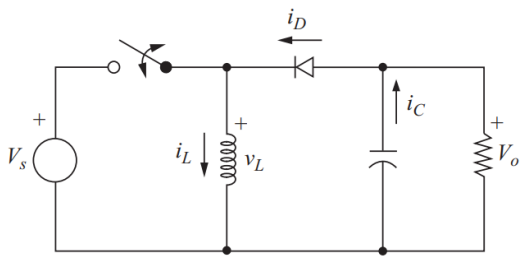

**(03)**

**1B.** Analyze the DC-DC converter's source side inductor current and switch current waveforms explaining the working.



**(03)**

**1C** Analyze the waveforms of inductor current, inductor voltage and capacitor current of the DC-DC converter given below. Derive an expression to estimate the output voltage.

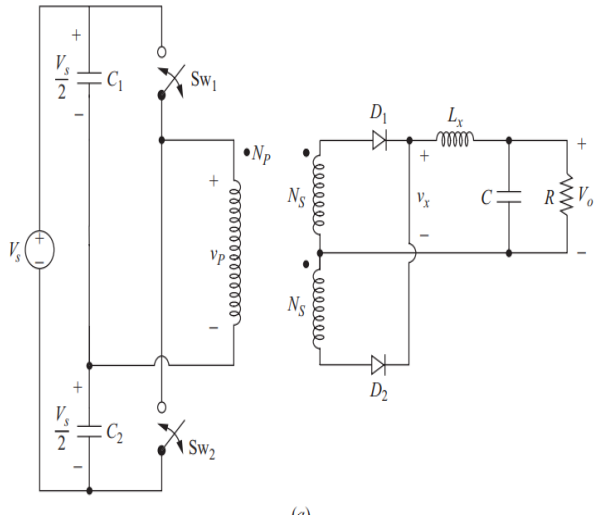


**(04)**

**2A.** A forward converter has the following parameters: source voltage 200V, (N1/N2) = (N1/N3) = 1, magnetizing inductance Lm 1mH, Lx is 70uH, load resistance 50Ω, capacitance 66uF and duty cycle 40%. Assume the switching frequency be 250 kHz. Determine (a) Output voltage (b) peak current through Lm in the transformer model (c) output ripple in percentage. **(04)**

**2B.** With the circuit diagram and working, analyze the inductor current on the load side of a push-pull converter. **(03)**

**2C.** Discuss the working of a DC-DC converter shown below and examine the voltage waveform $v_x$ . **(03)**




**3A.** Discuss any four factors affecting the choice of zero voltage switching resonant converters in switch mode power supply. **(02)**

**3B.** With the neat circuit diagram and waveform explain the working of parallel loaded resonant (PLR) converter with operating frequency below half the resonance frequency. **(04)**

**3C.** A series loaded resonant DC-DC converter has the following parameters: input voltage: 10V, resonant Inductor 6uH, resonant capacitor 6nF, switching frequency 900 kHz, load resistance 10Ω. Determine the output voltage, normalized switching frequency and quality factor. $V_0 = \frac{V_s}{2}\left(\frac{1}{\sqrt{1+\left[\frac{X_L - X_C}{R_e}\right]^2}}\right)$ **(04)**

**4A.** Design and explain the small signal ac model of the buck-boost converter into canonical form with step by step procedure. **(04)**

**4B.** Examine the need for modelling of a power converter with appropriate justifications. **(03)**

**4C.** The Zero Current Switching resonant converter has a source voltage of 12V, load resistance of 5Ω, switching frequency 100 kHz and output current is 1 A. Determine the values of inductor Lr and capacitor Cr in the resonant tank circuit such that its impedance Zr is 10 Ω and resonant frequency is 1 MHz. Also determine the peak current in Lr and peak voltage across Cr. **(03)**

**5A.** Ferrite core is more commonly used in the inductor design for power electronic converters. Justify. **(03)**

**5B.** Determine the inductance of the air-core coil with length 1m and radius 2mm, number of turns be 200 **(02)**

**5C.** Design an inductor for the buck converter with following specifications: Switching frequency is 35KHz, Output voltage = 8V, Input voltage = 12 V ± 10 %, Output Current, Io = 5A, Current ripple is 5% of Io, Bm = 0.2 T for ferrite, J = 4A/mm$^2$,Kc = 1.00

| Core | Mean length (mm) | Core cross-section (Ac*100mm$^2$) | Window area (Aw*100mm2) | Area Product (Ap*10^4 mm$^2$) | AL |
|---|---|---|---|---|---|
| EE20/10/5 | 38 | 0.31 | 0.478 | 0.149 | 1624 |
| EE25/9/6 | 51.2 | 0.40 | 0.78 | 0.312 | 1895 |
| EE25/13/7 | 52 | 0.55 | 0.87 | 0.478 | 2285 |
| EE30/15/7 | 56 | 0.597 | 1.19 | 0.71 | 3025 |
| EE36/18/11 | 70.6 | 1.31 | 1.41 | 1.847 | 4200 |
| EE42/21/9 | 77.6 | 1.07 | 2.56 | 2.739 | 2613 |
| EE42/21/15 | 93 | 1.82 | 2.56 | 4.659 | 4778 |
| EE42/21/20 | 99 | 2.35 | 2.56 | 6.016 | 6231 |
| EE65/32/13 | 150 | 2.66 | 5.37 | 14.284 | 4833 |

**(05)**